Solutions, CM2014, May 2022

1a, $x_k = x_{k-1} + h(-x_{k-1} t_{k-1} + 1)$, $x_0 = 1$
$t_k = t_{k-1} + h$, $t_0 = 0$

$h = 0.5$, $x_1 = x_0 + h(-x_0 t_0 + 1) = 1 + 0.5(0+1) = 1.5$
$t_1 = t_0 + h = 0.5$
$x_2 = x_1 + h(-x_1 t_1 + 1) = 1.5 + 0.5(-1.5 \cdot 0.5 + 1) = 1.625$
$t_2 = t_1 + h = 1$ $\quad x(1) \approx x_2 = 1.625$

b, $x_k = x_{k-1} + h(-x_k t_k + 1)$, $x_0 = 1$
$t_k = t_{k-1} + h$, $t_0 = 0$

$h = 0.5$, $x_1 = x_0 + h(-x_1 t_1 + 1)$
$t_1 = t_0 + h$
$x_1 = 1 + 0.5(-x_1 \cdot 0.5 + 1)$, $x_1 = \frac{1.5}{1.25} = 1.2$
$t_1 = 0.5$

2, a, $y = \cos(t)$, $\frac{dy}{dt} = -\sin t$

$H(t) = \cos^2(t) + \sin^2(t) = 1$ for all t

b, Let $u_1 = y$, $u_2 = \frac{dy}{dt}$

$\dot{u}_1 = u_2$
$\dot{u}_2 = -u_1$

$\dot{\bar{u}} = \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}}_{A} \bar{u}$, $\bar{u}(0) = \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\bar{u}_0}$

c) Euler's explicit method will <u>not</u> preserve the energy, see Lab 1

d) The IMM $\bar{u}_{k+1} = \bar{u}_k + \frac{h}{2}(A\bar{u}_{k+1} + A\bar{u}_k)$

$$\underbrace{\left(I - \frac{h}{2}A\right)}_{\begin{pmatrix} 1 & -\frac{h}{2} \\ \frac{h}{2} & 1 \end{pmatrix}} \bar{u}_{k+1} = \underbrace{\left(I + \frac{h}{2}A\right)}_{\begin{pmatrix} 1 & \frac{h}{2} \\ -\frac{h}{2} & 1 \end{pmatrix}} \bar{u}_k$$

$$\bar{u}_{k+1}^T \underbrace{\begin{pmatrix} 1 & \frac{h}{2} \\ -\frac{h}{2} & 1 \end{pmatrix}\begin{pmatrix} 1 & -\frac{h}{2} \\ \frac{h}{2} & 1 \end{pmatrix}}_{\begin{pmatrix} 1+\frac{h^2}{4} & 0 \\ 0 & 1+\frac{h^2}{4} \end{pmatrix}} \bar{u}_{k+1} = \bar{u}_k^T \underbrace{\begin{pmatrix} 1 & -\frac{h}{2} \\ \frac{h}{2} & 1 \end{pmatrix}\begin{pmatrix} 1 & \frac{h}{2} \\ -\frac{h}{2} & 1 \end{pmatrix}}_{\begin{pmatrix} 1+\frac{h^2}{4} & 0 \\ 0 & 1+\frac{h^2}{4} \end{pmatrix}} \bar{u}_k$$

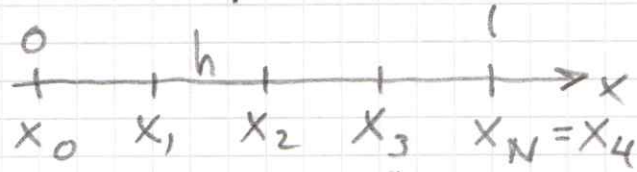
$$\bar{u}_{k+1}^T \left(1+\frac{h^2}{4}\right)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \bar{u}_{k+1} = \bar{u}_k^T \left(1+\frac{h^2}{4}\right)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \bar{u}_k$$

$$\bar{u}_{k+1}^T \bar{u}_{k+1} = \bar{u}_k^T \bar{u}_k = \bar{u}_0^T \bar{u}_0 = 1$$

3) a) Boundary Value Problem (BVP)

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + x_i y_i = 1, \quad y_0 = 1, \quad y_N = 0$$

$4h = 1, \quad h = 0.25$



$i = 1$: $y_2 - 2y_1 + \underbrace{y_0}_{=1} + h^2 x_1 y_1 = 1$

$$\left(-2 + \frac{1}{16}\cdot\frac{1}{4}\right) y_1 + y_2 = 0$$

$i = 2$: $y_3 - 2y_2 + y_1 + h^2 x_2 y_2 = 1$

$$y_1 + \left(-2 + \frac{1}{16}\cdot\frac{1}{2}\right) y_2 + y_3 = 1$$

$i = 3$: $\underbrace{y_4}_{=0} - 2y_3 + y_2 + h^2 x_3 y_3 = 1$

$$y_2 + \left(-2 + \frac{1}{16}\cdot\frac{3}{4}\right) y_3 = 1$$

$A\bar{y} = \bar{b}$

$$\begin{pmatrix} a & 1 & 0 \\ 1 & d & 1 \\ 0 & 1 & c \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$a = -2 + \frac{1}{64}$

$d = -2 + \frac{1}{32}$

$c = -2 + \frac{3}{64}$